Foundation Models, LLM, Embeddings

Week 1: Modern AI

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Session Topics

- ✔ What are Foundation Models?
- ✔ How Foundation Models are Trained
- ✔ Applications of Foundation Models
- ✔ What are Large Language Models?
- ✔ What Can LLMs Do?
- ✔ How LLMs Work & Generate Text
- ✔ What are Embeddings?
- ✔ How Embeddings Represent Meaning
- ✔ Applications of Embedding

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What are Foundation Models?

- A Foundation Model is a large AI model trained on broad and diverse datasets so that it can perform a wide range of tasks.
- Instead of building a separate model for every individual task, a foundation model can be adapted to many different applications.
- These models learn general patterns, representations, relationships, and structures from large amounts of data.
- They can process different types of information, including text, images, audio, video, and code.
- Foundation models form the technological base for many modern Generative AI applications.

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What are Foundation Models?

Key Characteristics

- **Large-scale training** → Trained using very large and diverse datasets.
- **General-purpose capability** → Can support multiple tasks rather than one specific task.
- **Transferability** → Knowledge learned during training can be adapted to new tasks.
- **Adaptability** → Can be customized through prompting, fine-tuning, or other techniques.
- **Multimodal capability** → Some modern models can work with multiple types of data.

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How are Foundation Models Trained?

Foundation models are developed through large-scale training that allows the model to learn general patterns from massive datasets.

Data Collection

- Large and diverse datasets are collected from sources such as text, images, code, audio, and other digital content.
- The quality and diversity of the training data strongly influence the capabilities of the resulting model.

Data Preparation

- Data is cleaned, filtered, deduplicated, and transformed into a format suitable for training.
- Sensitive, harmful, low-quality, or irrelevant data may be filtered depending on the training process.

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How are Foundation Models Trained?

Pretraining

- The model is trained on large amounts of data using a learning objective.
- For language models, a common objective is predicting missing or upcoming tokens.
- Through repeated training, the model learns patterns such as language structure, relationships, concepts, and associations.

Model Learning

- The model contains a large number of parameters that are adjusted during training.
- These parameters capture patterns learned from the training data.
- Training typically requires significant computing resources and specialized hardware.

Adaptation

- After pretraining, the model can be adapted for specific uses through techniques such as: Fine-tuning, Instruction tuning, Reinforcement learning from feedback and Prompting

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Applications of Foundation Models

Foundation models can serve as a common AI foundation for many professional applications.

- **Content Generation:** Generate articles, reports, emails, summaries, and marketing content.
- **Code Generation:** Generate, explain, document, and assist with software code.
- **Knowledge Assistance:** Answer questions and help users interact with organizational information.
- **Document Processing:** Summarize documents, extract information, classify content, and compare documents.
- **Multimodal Applications:** Some models can process combinations of text, images, audio, and other data types.

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Applications of Foundation Models

Business Applications

- **Customer Support** → AI assistants and automated responses.
- **Marketing** → Content creation and customer analysis.
- **Finance** → Document analysis and information extraction.
- **Healthcare** → Information summarization and decision support.
- **Software Development** → Coding assistance and documentation.
- **Education** → Personalized learning and content generation.

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What are Large Language Models (LLMs)?

- A Large Language Model (LLM) is an AI model trained on very large amounts of text and other data to understand and generate human-like language.
- LLMs are large-scale language models capable of understanding and generating text across a wide range of tasks.
- Most modern LLMs are based on the Transformer architecture.
- LLMs learn statistical relationships between tokens and use those relationships to generate or predict text.
- They can perform multiple language tasks without requiring a separate model for each task.
- LLMs do not simply store complete answers like a traditional database. They learn patterns from training data and generate responses based on the context provided to them.

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What Can LLMs Do?

- **Answer Questions** → Generate responses to user queries.
- **Summarize** → Convert long documents into concise summaries.
- **Translate** → Convert text between languages.
- **Generate Content** → Create emails, reports, stories, and other text.
- **Analyze Text** → Identify sentiment, topics, entities, and patterns.
- **Generate Code** → Create and explain programming code.
- **Reason Over Information** → Combine information to solve many structured problems.

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How Do LLMs Work & Generate Text?

LLMs generate text by processing input as tokens and predicting what should come next.

Step 1 — User Input:

- The user provides a prompt or instruction.

Step 2 — Tokenization

- The input is divided into smaller units called tokens.
- Tokens may represent complete words, parts of words, punctuation, or special symbols.

Step 3 — Embedding

- Tokens are converted into numerical vectors called embeddings.
- These vectors allow the model to process the input mathematically.

Step 4 — Transformer Processing

- The Transformer uses self-attention to determine relationships between tokens.
- Multiple Transformer layers process the information and build contextual representations.

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How Do LLMs Work & Generate Text?

Step 5 — Next-Token Prediction

- The model calculates probabilities for possible next tokens.
- A token is selected based on the model's probability distribution and the generation settings.

Step 6 — Repeat

- The newly selected token is added to the context.
- The model predicts the next token again.
- This process continues until the response is complete.

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What are Embeddings?

- Embeddings are numerical representations of data that capture meaningful relationships between different items.
- Words, sentences, images, documents, or other objects can be converted into vectors in a mathematical space.
- Similar or related items tend to have representations that are closer together in this space.
- Embeddings allow AI systems to work with meaning and relationships mathematically.

Why are Embeddings Important?

- Computers cannot directly understand the meaning of a sentence in the same way humans do. Embeddings convert information into numerical representations that AI systems can compare, search, and process.

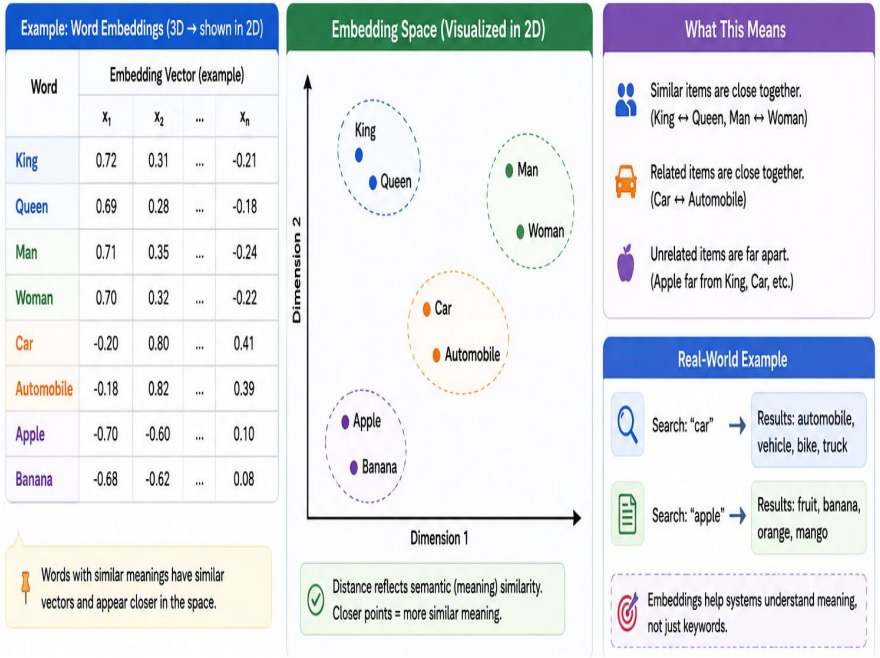
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How Embeddings Represent Meaning

- Embeddings place related items close together in a **vector space**.
- For example:
- **“Car”** and **“Automobile”** likely have similar representations.
- **“Car”** and **“Banana”** likely have less similar representations.
- The similarity between embeddings can be measured using mathematical techniques such as cosine similarity.



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Applications of Embeddings

- **Semantic Search:** Finds information based on meaning rather than exact keyword matching.
- **Recommendation Systems:** Identifies products, content, or services that are similar to a user's interests.
- **Document Similarity:** Compares documents based on their semantic meaning.
- **Clustering:** Groups similar documents, customers, products, or other data points.
- **Classification:** Embeddings can be used as input features for classification models.

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